

TZS-MAG-1600-B-User Manual

TZBOT
zhejiang Tongzhu technology Co., Ltd.

Function

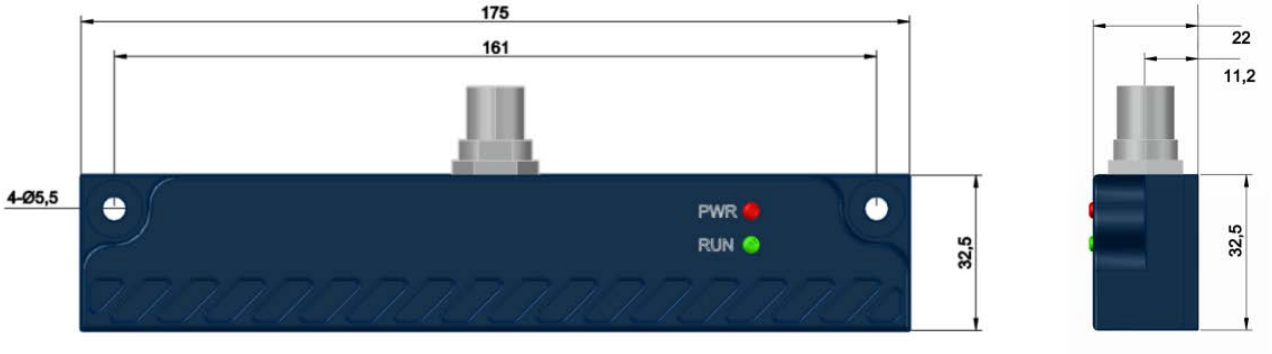
The TZS-MAG-1600-B can detect the intensity distribution of the magnetic field along the horizontal direction and accurately calculate the deviation between the center of the magnetic field by high-order fitting. The deviation information can be use to guide the AGV.

Property

Model		TZS-MAG-1600-B
Shell material		ABS+PC or aluminum alloy
Dimensions		L175mm xW40mm x H21.5mm
Weight		Below 300g
Detect scope		160mm
Detect schedule		10~70mm
Detection accuracy		1mm
Segmentation		0.1mm
Feedback cycle		4ms、10ms、20ms、100ms
Applicable tape width		30mm, 50mm
Detect polarity		N/S
Fork identification number		Left, middle, right
Supply voltage	Input voltage	DC24V (DC9~36V)
	Input current	Working current: <200mA
Communication Interface	CAN	CAN custom data, data transmission rate: 0.125M ~ 1M bit/s
	RS485	Standard Modbus, data transmission rate: 9600bps ~ 115200bps
	RS232	Serial communication, data transmission rate: 9600bps ~ 115200bps
Indicator light	PWR	Power indicator light, red, always on when powered on
	RUN	Running indicator light, green, blinks when running, always on when magnetic stripe is detected
Protection level	IP67	Able to adapt to harsh environments such as oil (cutting fluid, etc.), dust, and humidity
Environmental conditions	Temperature	Operating temperature -25°C~85°C (no freezing)
	Humidity	20~95%RH or less during use and storage (no condensation)
Certification standards		CE
Appendix		Cable: M12S-8P150 (standard)

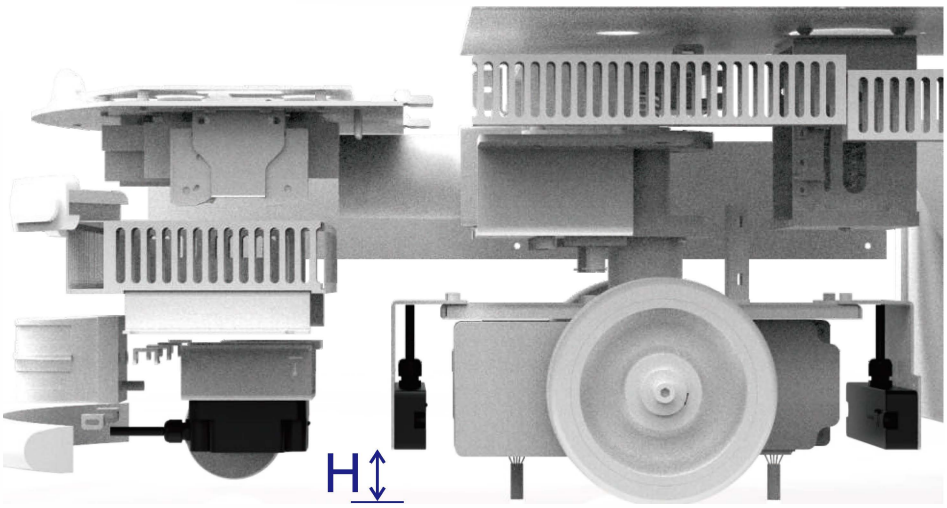
Installation

TZS-MAG-1600-B



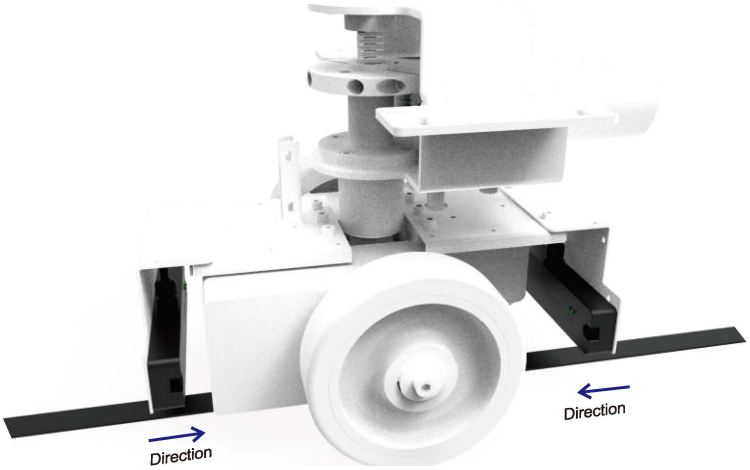
Note: Use M5 screws to fix.

Mount H



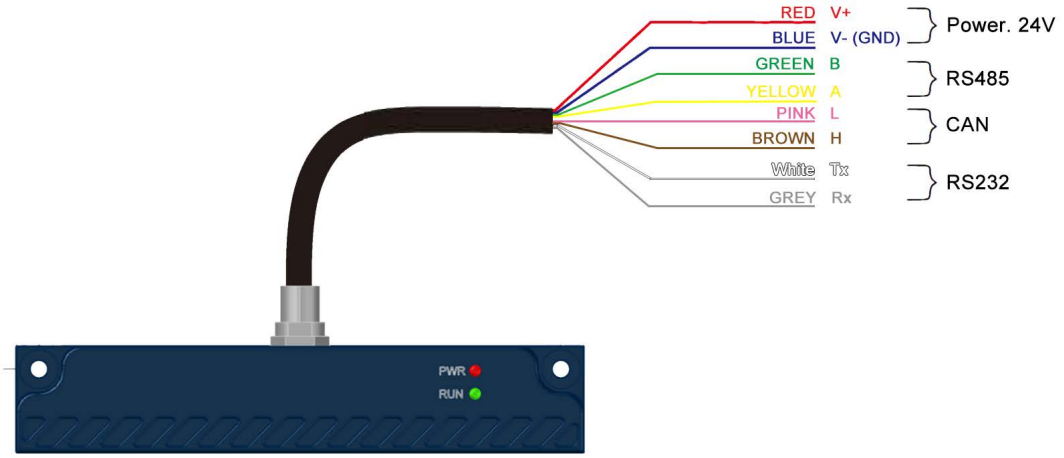
The bottom surface to the ground distance H should be around 35mm for best performance.

Mount Direction



Standard Mount Direction should be Above. The front and rear sensor Direction should be opposite.You can change your mount direction, But you need to configure it right in AGVStudio.

Pin Assignment



TZS-MAG-1600-B offer 3ways to obtain the Sensor data, Rs485, RS232 and Can.

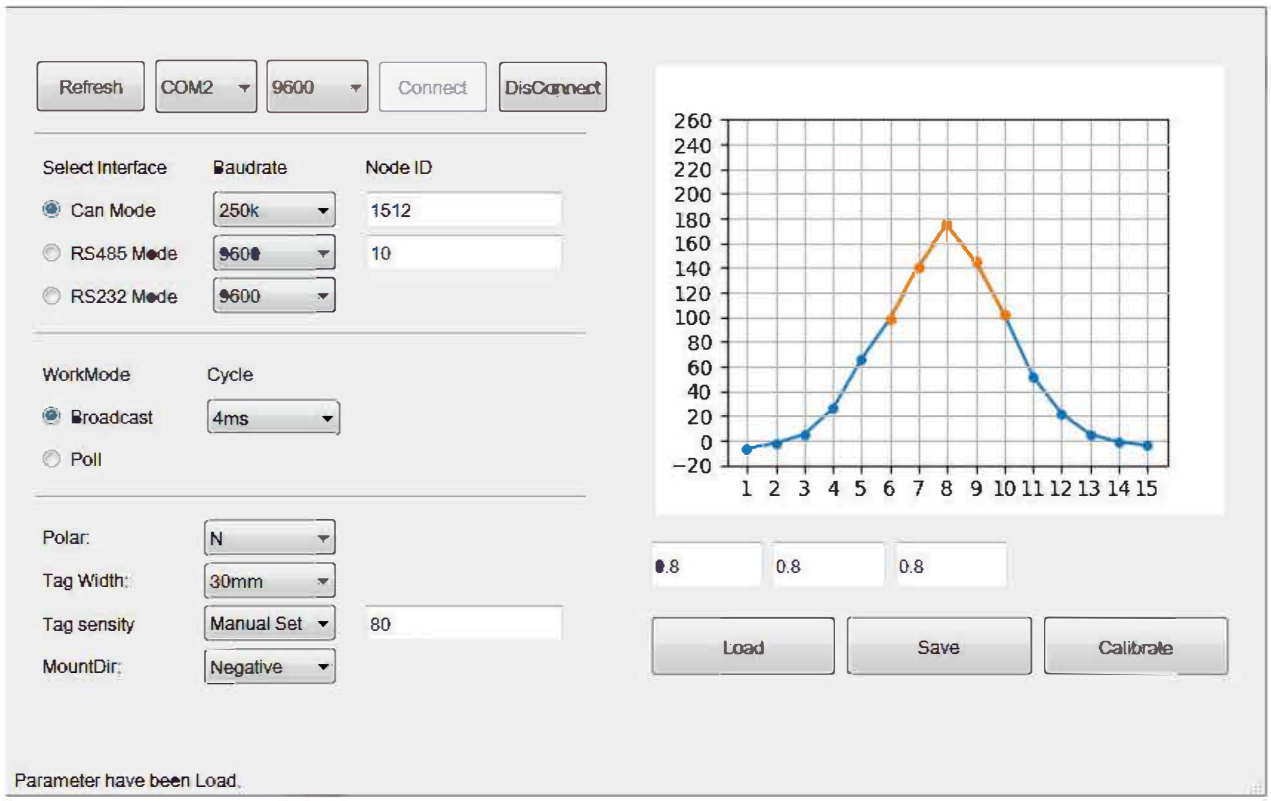
Software

Use the‘MagtoolCom.exe’ to configure the sensor. MagToolCom is a free-installation software as below.

Software ICON

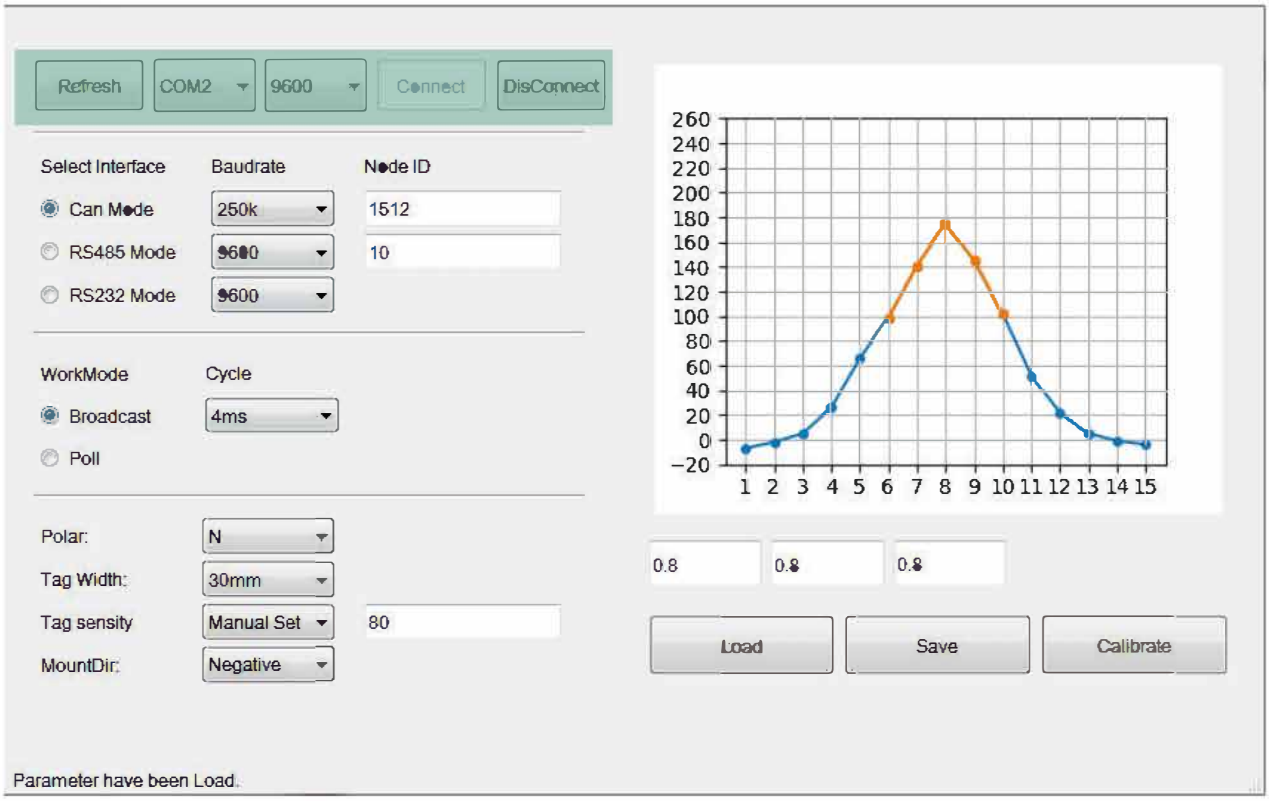


Software Interface



Software

Connection



We can connect to the sensor by Rs485 or Rs232. The default communicate parameter is as below

data bit : 8
parity bit : 1
parity mode : None
baud rate : 9600bps

Check your connect diagram if you can not connect to the sensor.



RS485 connection

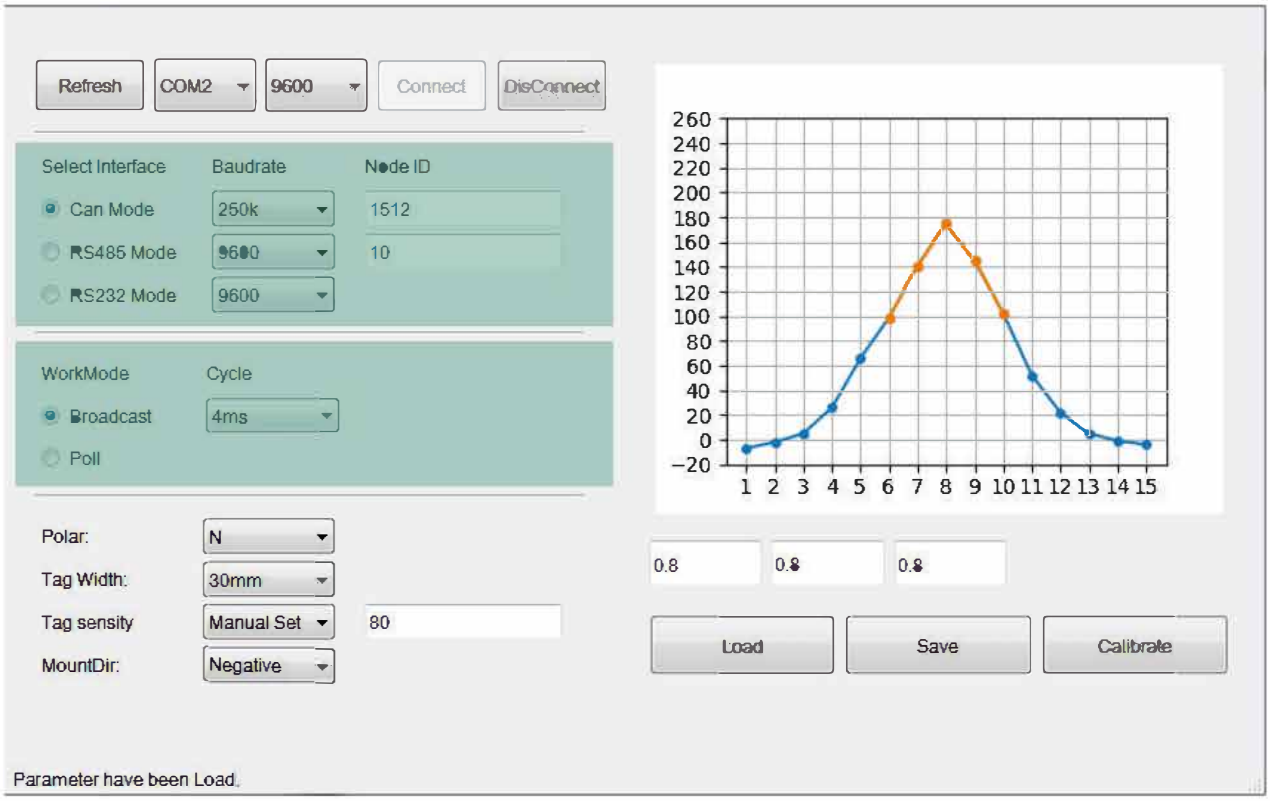


RS232 connection

Note: You need to 'save' the configuration to the sensor by pressing Save button. You can also read out the configuration out of the sensor.

Software

Interface & work mode

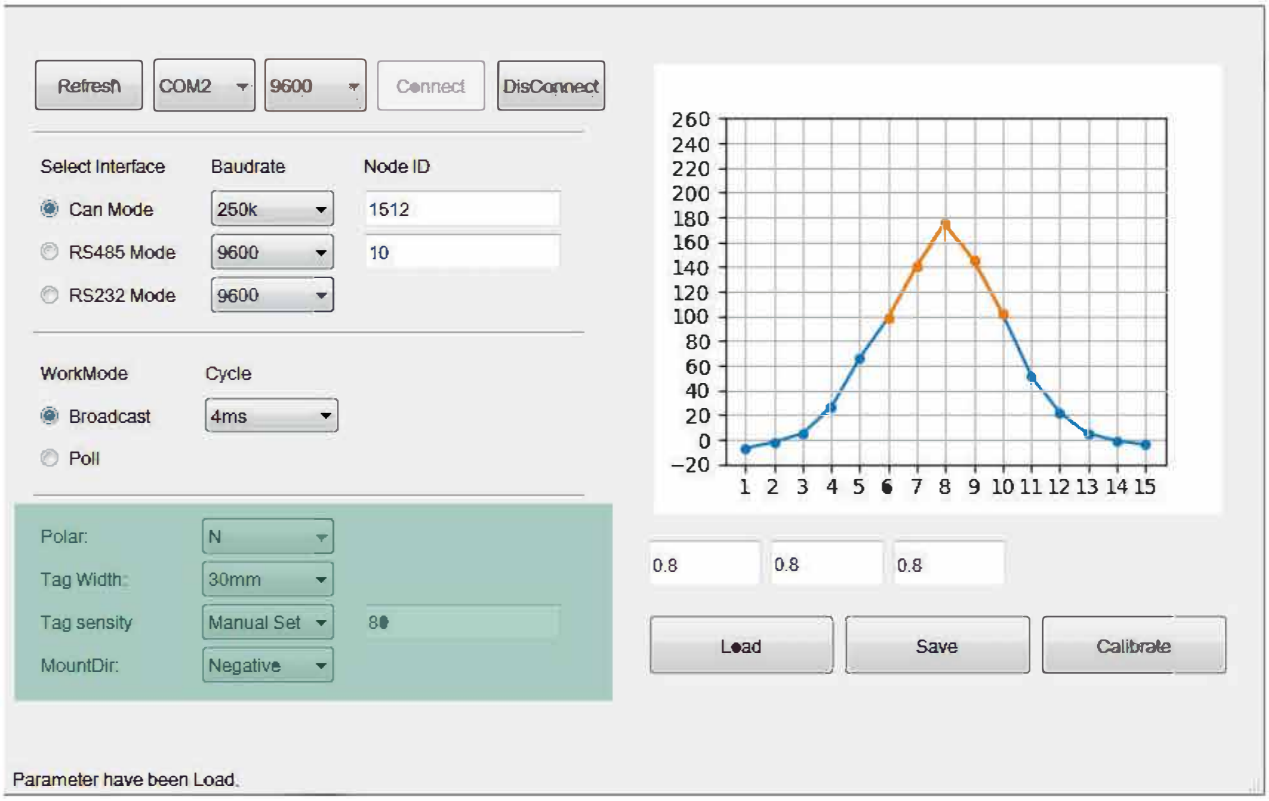


As what we show above. You can configure the sensor work mode to broadcast or Poll by Master. The respond interface can be one of Can or Rs485.

Note: You need to 'save' the configuration to the sensor by pressing Save button. You can also read out the configuration out of the sensor.

Software

Sensor key parameter

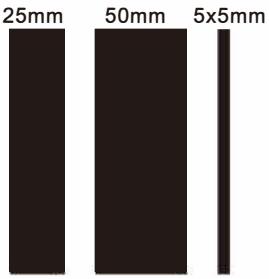


Polar:



Depends on which type of magnetic taps you use.

Tap Width:

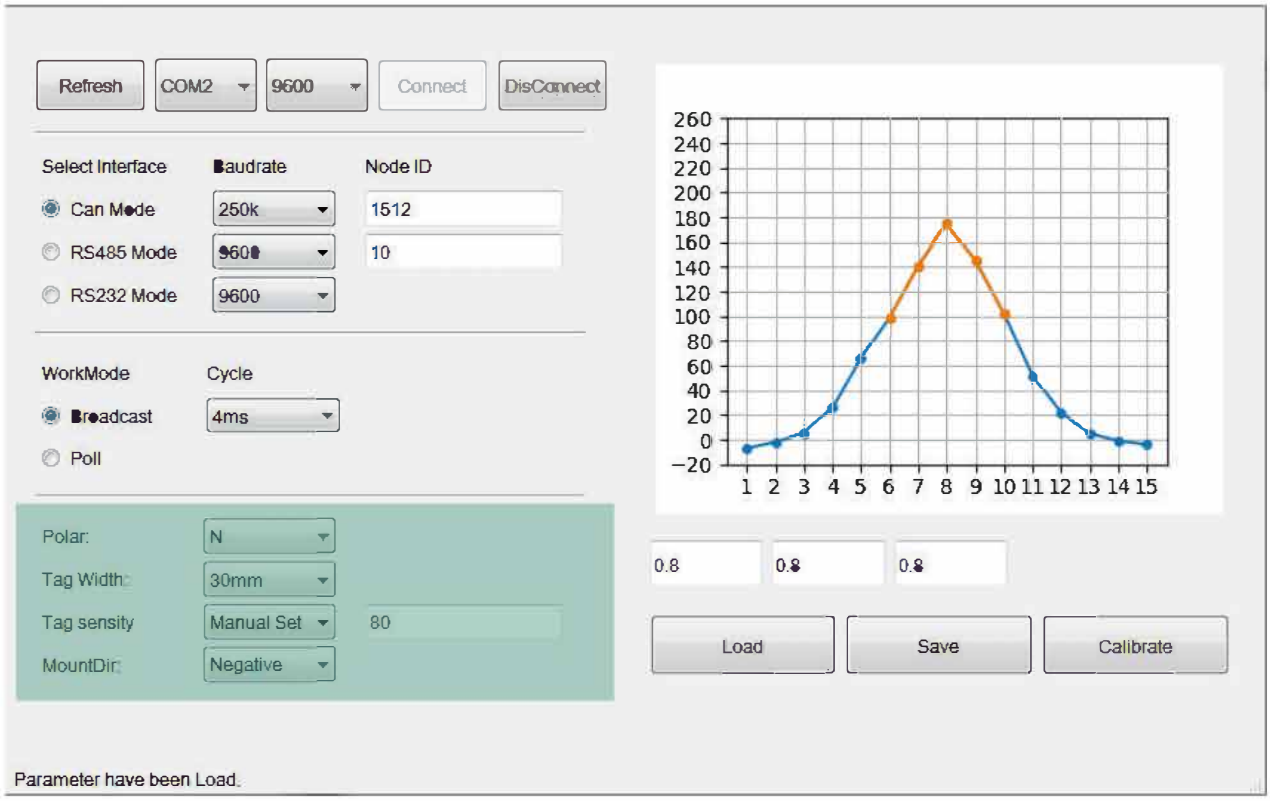


Depends on which type of magnetic taps you use.

Note: You need to 'save' the configuration to the sensor by pressing Save button. You can also read out the configuration out of the sensor.

Software

Sensor key parameter

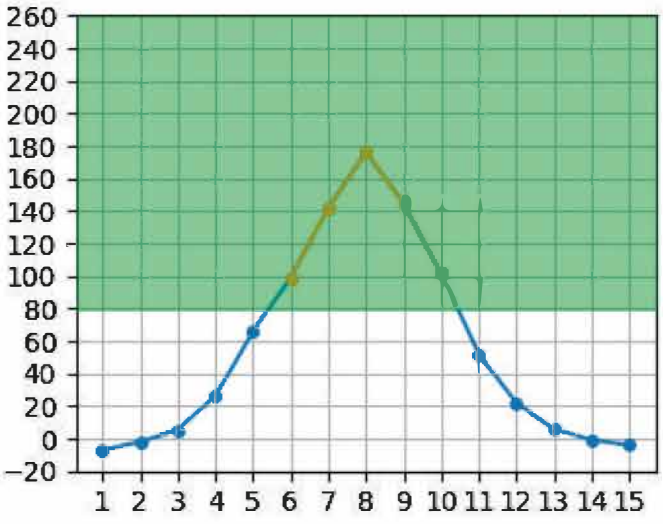


Tap Sensity:

The sensor use this value to recognize if there is Taps occurs when the actual density is greater than the configure value.

When you configure the value to be 80, Because the density of sensor cell (total 15 cells) 6~10 are greater than 80 and they will light up in the figure.

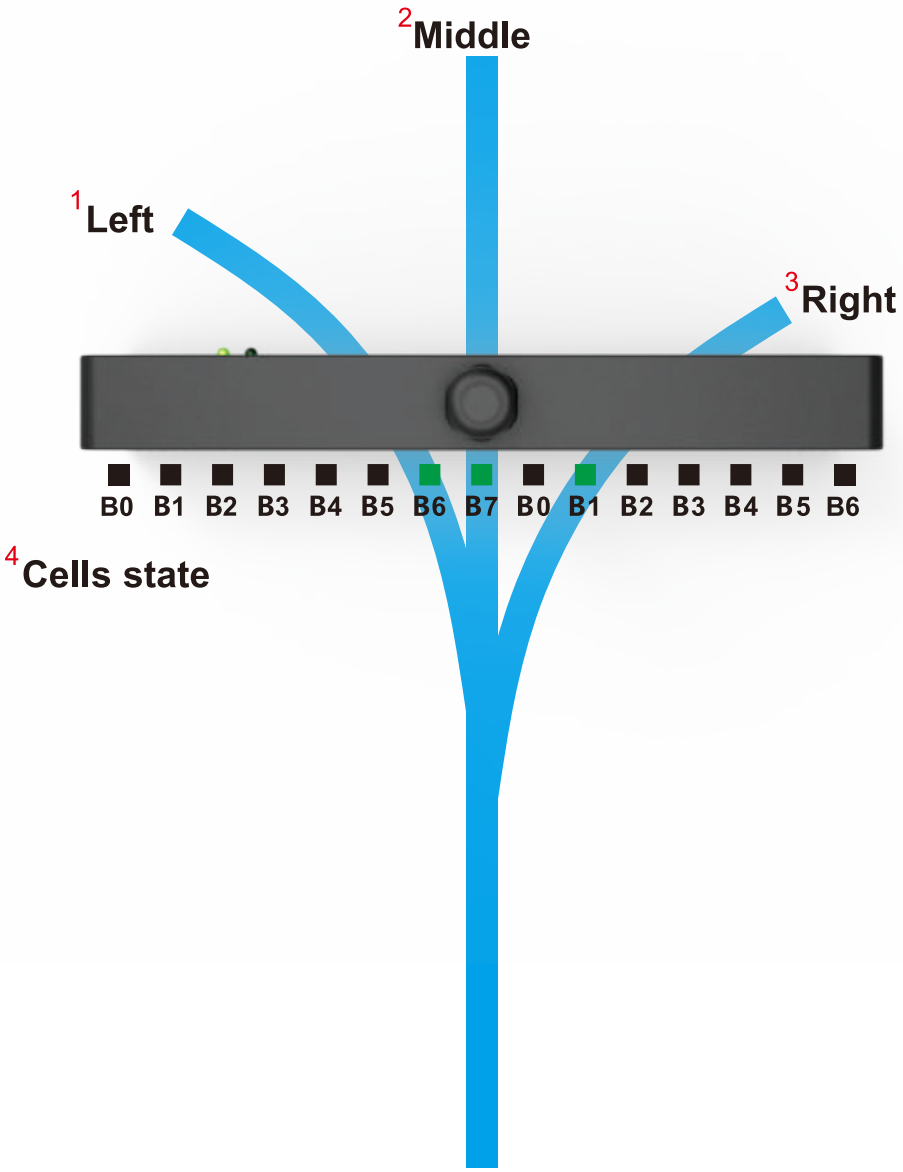
To note that the interval of the cell is 10mm. So you need to make sure the value you configure is reasonable. For example: If the tap is 30mm width. So cell 10 - cell 6 = 4 interval(within 40mm) is reasonable.



Note: You need to 'save' the configuration to the sensor by pressing Save button. You can also read out the configuration out of the sensor.

CanData

Data on sensor



1. Left - The left fork Position;
2. Middle - The middle fork Position;
3. Right - The right fork Position;
4. Cells state - The sensor-cells'status, depends on the Tag sensity you configure.

CanData

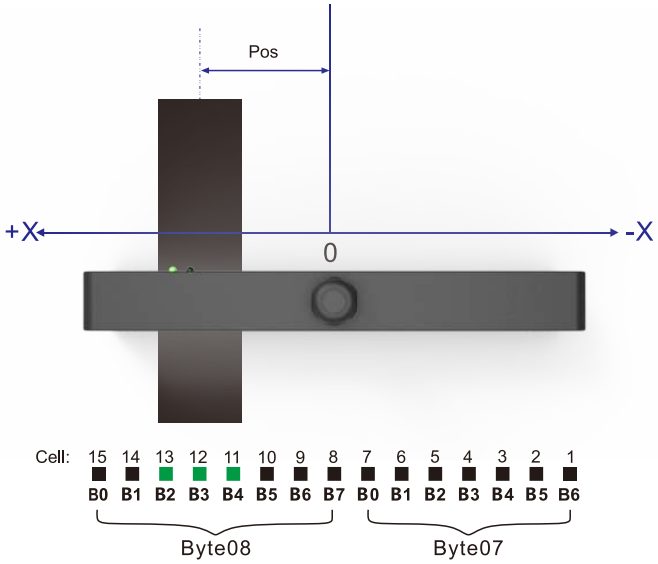
Can Data payload

There are 8 bytes for the data. Can frame : DLC=8, RTR=DATA, IDE=STANDARD.

Byte01	Byte02	Byte03	Byte04	Byte05	Byte06	Byte07	Byte08
H	L	H	L	H	L	H	L
[3]Right		[2]Middle		[1]Left		[4]Cells state	

Pos data

It is 0.1mm/unit. For example 200 means the deviation is 20mm, when there is no taps detected, the data respond would be 0x8000;
Be care of the signs of the data. It base on the coordinate below.



Note : When there are no taps , all 3 pos data value is 0x8000. When only one fork, all 3 pos data are the same. When two forks, right and middle data are the same.

Cells state

There are 15 cells in one sensor. So there total 15bits to express their states. As what you see above.

1. Left - The left fork Position;
2. Middle - The middle fork Position;
3. Right - The right fork Position;
4. Cells state - The sensor-cells'status, depends on the Tag sensity you configure.

Rs485 Data

Rs485 Data payload - BroadCast

There are 14 bytes for the data.

Byte01	Byte02	Byte03	Byte04	Byte05	Byte06 ~ Byte13	Byte14
0xAA	0x53	0x0B	0x01	NodeID	MagDatas	0xAE

Byte06	Byte07	Byte08	Byte09	Byte10	Byte11	Byte12	Byte13
H	L	H	L	H	L	H	L
[3]Right		[2]Middle		[1]Left		[4]Cells state	

Poll Mode - ModBus

Communicate parameters.

data bit : 8
parity bit : 1
parity mode : None
baud rate : 9600 / 19200 / 38400 bps

ModBus Function Supported: 04H
Address Range : 1000 ~ 1010 (decimal)

DATA:

1000	1001	1002	1003	1004~1010
[3]Right	[2]Middle	[1]Left	[4]Cells state	Reserved

1. Left - The left fork Position;
2. Middle - The middle fork Position;
3. Right - The right fork Position;
4. Cells state - The sensor-cells'status, depends on the Tag sensity you configure.

Rs232 Data

Rs232 Data payload - BroadCast

There are 44 bytes for the data.

Byte01	Byte02	Byte03	Byte04	Byte05	Byte06 ~ Byte35	Byte36~Byte43	Byte44
0xAA	0x53	0x29	0x01	0x00	Reserve	MagDatas	0xAE

Byte36	Byte37	Byte38	Byte39	Byte40	Byte41	Byte42	Byte43
H	L	H	L	H	L	H	L
[4]Cells state		[3]Right		[2]Middle		[1]Left	

Poll Mode

Request data by define data frame。

Request Data：

Byte01	Byte02	Byte03	Byte04	Byte05	Byte06	Byte07
0xAA	0x57	0x04	0x00	0x00	0x01	0xA1

Respond data:

The same as the broadcast data frame.

1. Left - The left fork Position;
2. Middle - The middle fork Position;
3. Right - The right fork Position;
4. Cells state - The sensor-cells'status, depends on the Tag sensity you configure.